Conditional Probability

Chapter Overview

1: Set Notation

How sets are used to describe a sample space/event and how notation like $A \cap B$ is used to combine sets.

2: Conditional Probability in Venn Diagrams

The notation $P(A|B)$ means "the probability of A given that B happened". How we can find such probabilities using a Venn Diagram.

3: Formula for Conditional

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

4: Tree Diagrams

"I have 3 red and 4 green balls in a bag. I take one ball out the bag, keep it, then take another. **Given that** the second ball was green, determine the probability the first was red."

RECAP :: Using sets for sample spaces and events

Solution 3

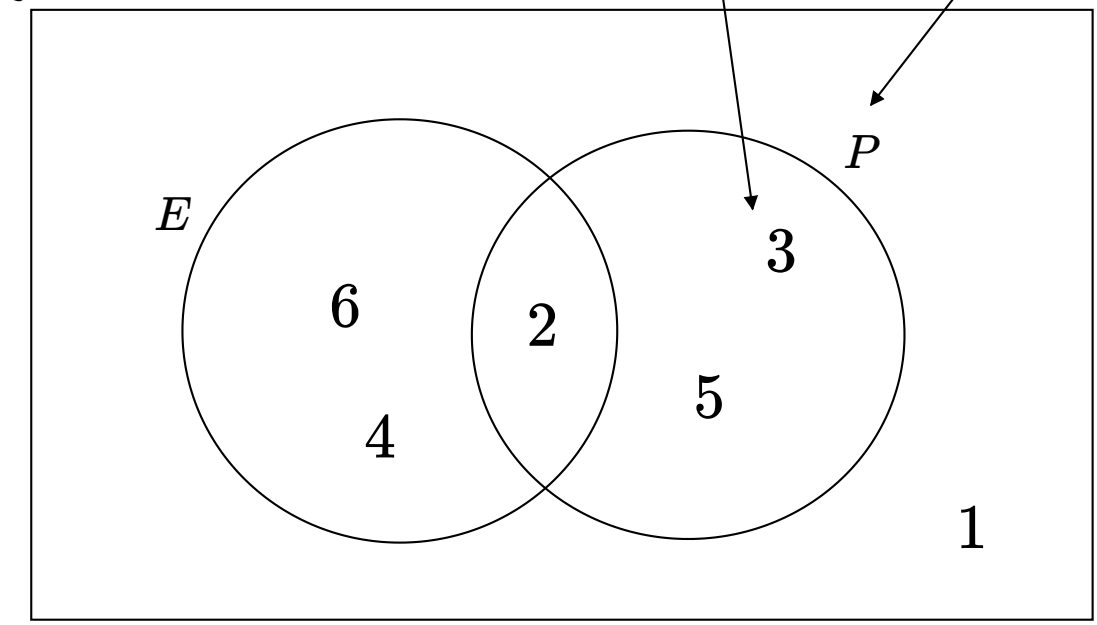
In general, sets are used to represent **collections of items**.

A **sample space** is **set of all possible outcomes**. We use ξ (Greek 'Xi'), or sometimes just S, to represent this set. We use a rectangle in a Venn Diagram.

Each number represents an **outcome**.

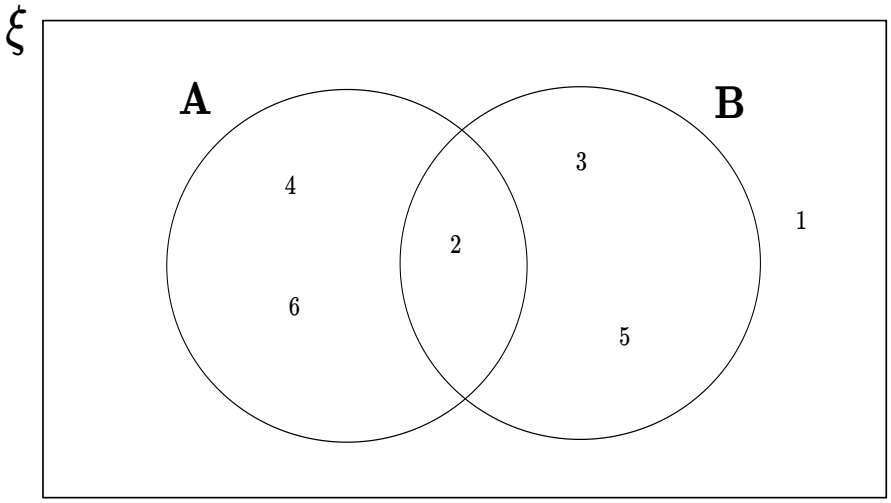
In probability, an **event** is a **set of one or more outcomes**. These are the circles in the Venn Diagram. We use capital letters for the variables representing sets.

E = "rolling an even number"
P = "rolling a prime number"



Combining events/sets

Solution 4



ξ = the whole sample space (1 to 6)

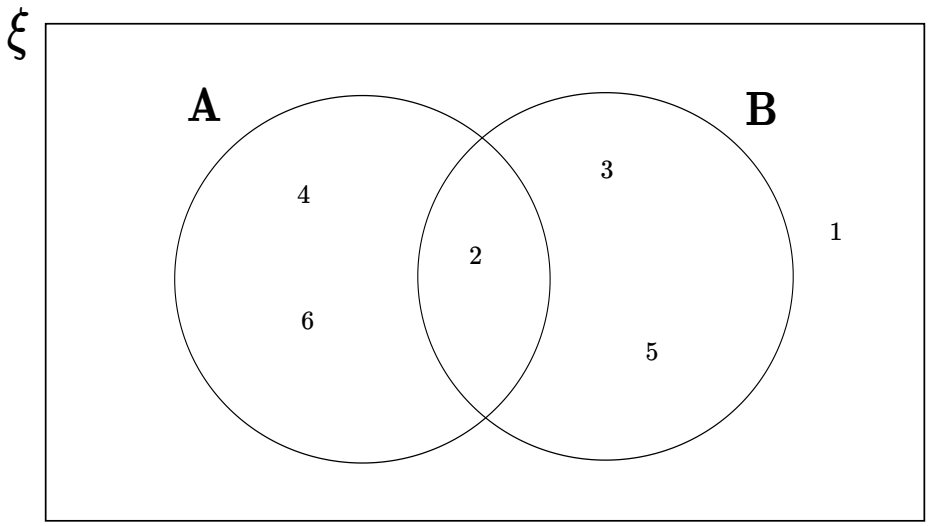
A = even number on a die thrown

B = prime number on a die thrown

	What does it mean in this context?	What is the resulting set of outcomes?
A'	**Not A (the "complement" of A).** i.e. Not rolling an even number.	$\{1, 3, 5\}$
$A \cup B$	**A or B (the "union" of A and B).** i.e. Rolling an even or prime number.	$\{2, 3, 4, 5, 6\}$
$A \cap B$	**A and B (the "intersection" of A and B).** i.e. Rolling a number which is even and prime.	$\{2\}$

Some fundamentals

Solution 5



S = the whole sample space (1 to 6)

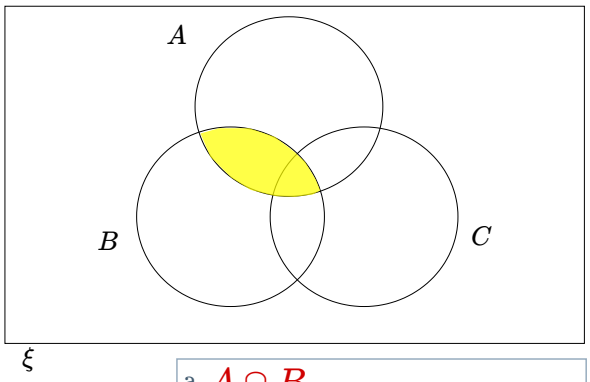
A = even number on a die thrown

B = prime number on a die thrown

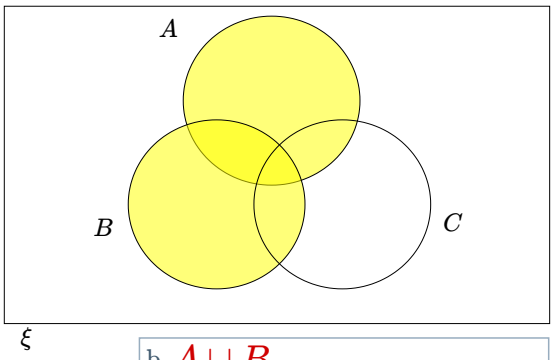
	What does it mean in this context?	What is the resulting set of outcomes?
$A \cap B'$	**"A and not B".** Rolling a number which is even and not prime.	$\{4, 6\}$
$(A \cup B)'$	Rolling a number which is not [even or prime].	$\{1\}$
$(A \cap B)'$	Rolling a number which is not [even and prime].	$\{1, 3, 4, 5, 6\}$

What area is indicated?

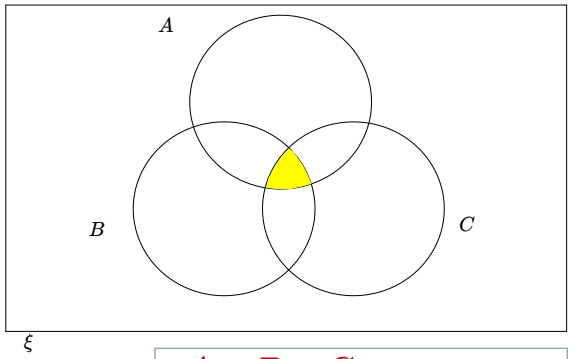
Solution 6



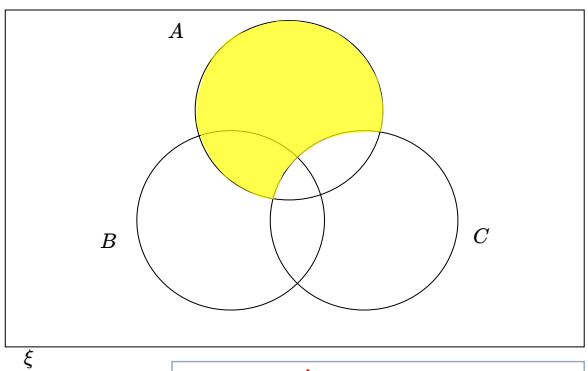
a $A \cap B$



b $A \cup B$



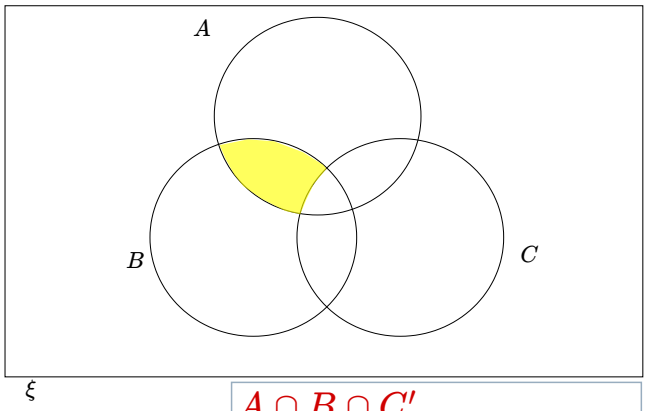
c $A \cap B \cap C$



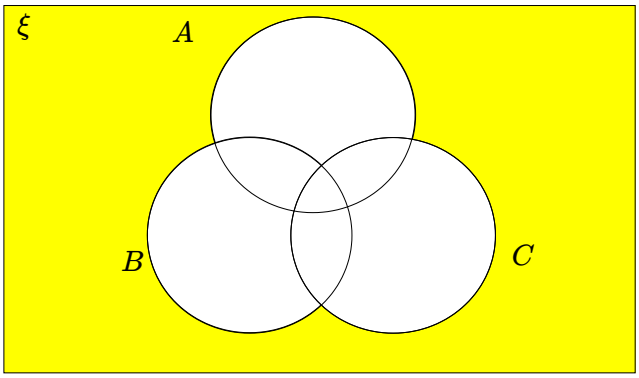
d $A \cap C'$

What area is indicated?

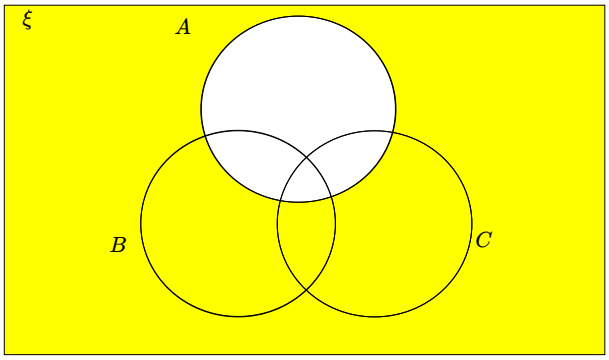
Solution 7



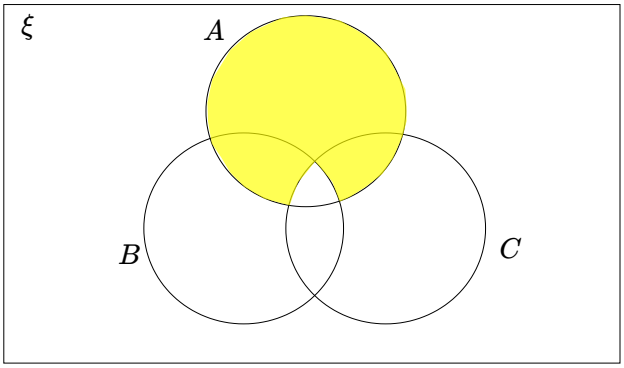
$A \cap B \cap C'$



$A' \cap B' \cap C'$ or $(A \cup B \cup C)'$



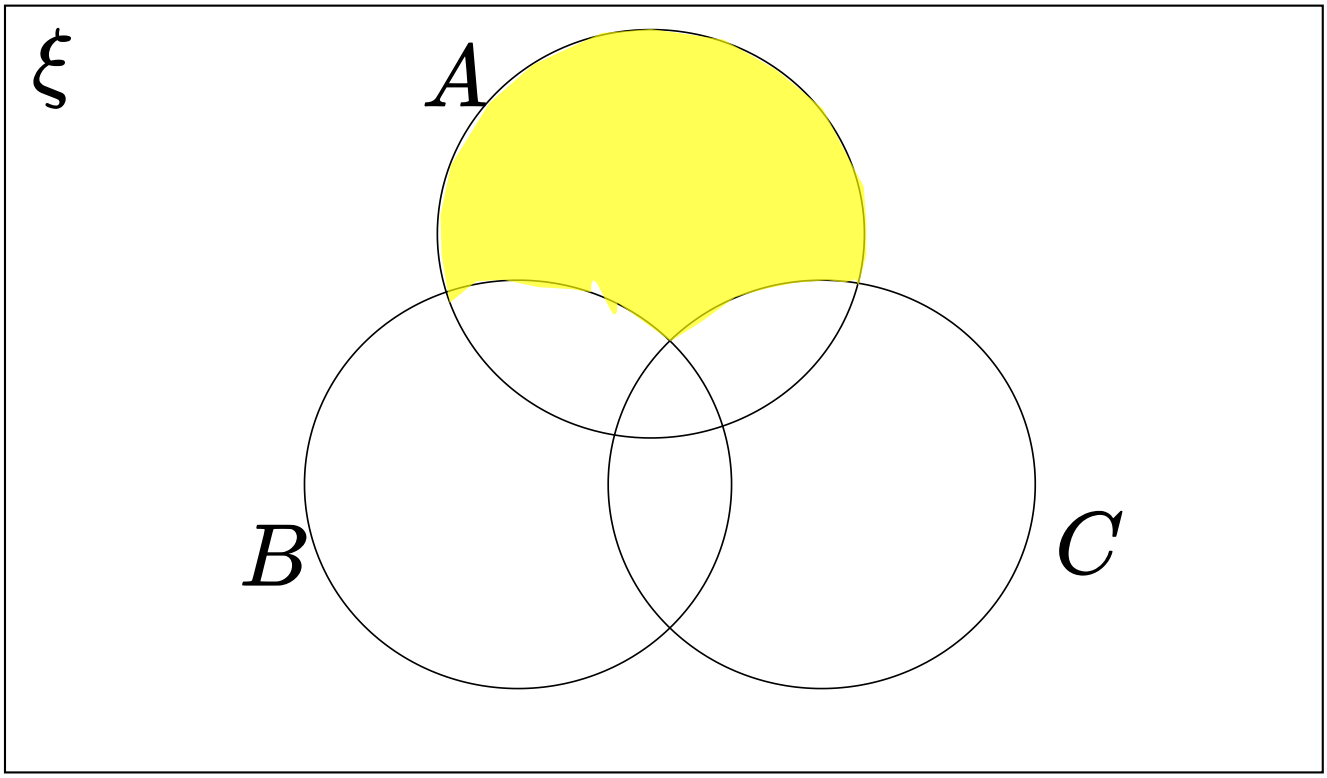
A'



$A \cap (B \cap C)'$

What area is indicated?

Solution 8



$A \cap B' \cap C'$

Examples

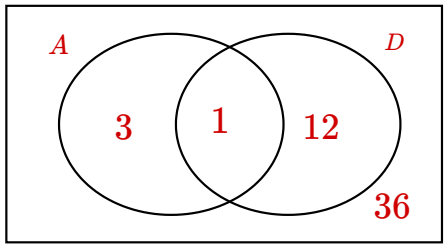
Solution 9

Venn Diagram can either contain:
(a) The **specific outcomes** in each set
(b) The number of items in the set (i.e. **frequencies**)
(c) The **probability** of being in that set.

This will usually be stated or made obvious from the context.

[Textbook] A card is selected at random from a pack of 52 playing cards. Let A be the event that the card is an ace and D the event that the card is a diamond. Find:
a) $P(A \cap D)$ b) $P(A \cup D)$ c) $P(A')$ d) $P(A' \cap D)$

Venn diagram



a $P(A \cap D) = \frac{1}{52}$

b $P(A \cup D) = \frac{16}{52}$

c $P(A') = \frac{48}{52}$

d $P(A' \cap D) = \frac{12}{52}$

Think "A and D".

For union, I visualise this 'figure-of-8' shape:



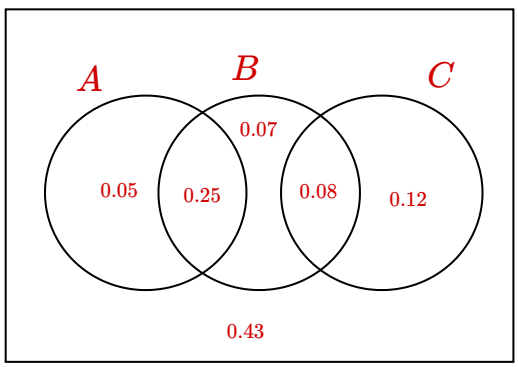
Examples

Solution 10

[Textbook] Given that $P(A) = 0.3$, $P(B) = 0.4$ and $P(A \cap B) = 0.25$,
a. Explain why events A and B are not independent.
Given also that $P(C) = 0.2$, that events A and C are mutually exclusive and that events B and C are independent,
b. Draw a Venn diagram to illustrate the events A, B and C, showing the probabilities for each region.
c. Find $P((A \cap B') \cup C)$

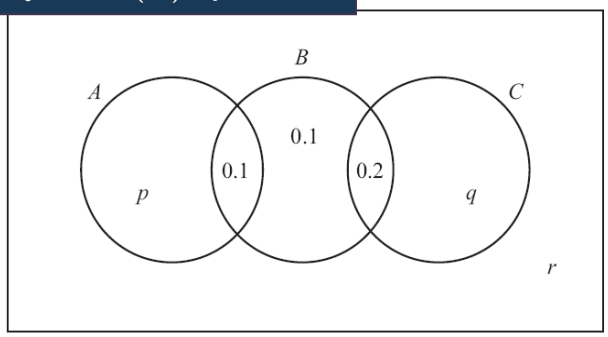
a $P(A)P(B) = 0.12 \neq 0.25 = P(A \cap B)$.

b



c
$P(A \cap B') = 0.05$
$P((A \cap B') \cup C) = 0.05 + 0.2 = 0.25$

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The Venn diagram in Figure 1 shows three events A, B and C and the probabilities associated with each region of B. The constants p, q and r each represent probabilities associated with the three separate regions outside B.

The events A and B are independent.

(a) Find the value of p. **(3)**

(a) Using the above:

$0.1 = (p + 0.1) \times 0.4$

$p + 0.1 = 0.25$

$p = 0.15$

Exercise 2A – Statistics Y2 Pages 19-21

9. $P(A) = 0.55$, $P(B) = 0.35$, $P(C) = 0.4$ and $P(A \cap C) = 0.2$. Events A, B are mutually exclusive; B, C are independent.

(a) Draw a Venn diagram. (4)

(b) Find each probability. (1 each)

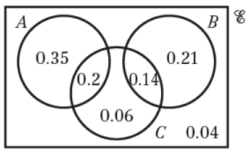
(i) $P(A' \cap B')$

(ii) $P(A \cup (B \cap C'))$

(iii) $P((A \cap C)' \cup B')$

a

$P(A \cap B) = 0$ and $P(B \cap C) = 0.35(0.4) = 0.14$. The regions only in A, B, C are respectively $0.35, 0.21, 0.06$. The outside region has probability $1 - 0.96 = 0.04$.



b(i) Outside A and B: $0.06 + 0.04 = 0.10$.

b(ii)

$0.55 + 0.21 = 0.76$ since the regions are disjoint.

b(iii)

The complement of $A \cap C$ omits only that intersection, which lies in B'. The required probability is 1.

Exercise 2A – Statistics Y2 Pages 19-21

10. $P(A) = 0.25$, $P(B) = 0.4$, $P(C) = 0.45$, $P(A \cap B \cap C) = 0.1$. The pairs A, B and B, C are independent, and $A \cap B' \cap C = \varnothing$.

(a) Draw a Venn diagram. (4)

(b) Find (i) $P(A' \cap (B' \cup C))$, (ii) $P((A \cup B) \cap C)$. (1 each)

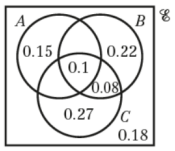
(c) Are A' and C independent? Justify. (2)

The empty set has probability zero.

Solution 13

a

$P(A \cap B) = 0.25(0.4) = 0.1$ and $P(B \cap C) = 0.4(0.45) = 0.18$. Subtract the triple intersection: the pair-only regions are $0, 0, 0.08$. The single-only regions are $0.15, 0.22, 0.27$ and the outside region is 0.18.



b(i)

$0.27 + 0.08 + 0.18 = 0.53.$

b(ii)

$0 + 0.10 + 0.08 = 0.18.$

c

$P(A')P(C) = 0.75(0.45) = 0.3375$, while $P(A' \cap C) = 0.08 + 0.27 = 0.35$. They are dependent.

Exercise 2A – Statistics Y2 Pages 19-21

11. A book club reads murder mysteries M, ghost stories G or epic fiction E. $P(M) = 0.5$, $P(G) = 0.4$, $P(E) = 0.6$. No member reads both ghost stories and epic fiction; $P(M \cap G) = 0.3$.

(a) Draw a Venn diagram. (4)

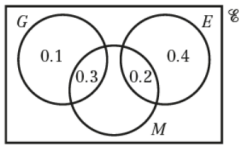
(b) Find (i) $P(M \cup G)$, (ii) $P((M \cap G) \cup (M \cap E))$. (2)

(c) Are G' and M independent? Justify. (2)

Solution 14

a

$P(G \cap E) = 0$. The G-only region is $0.4 - 0.3 = 0.1$. Every member outside G reads E; $P(M \cap E) = 0.5 - 0.3 = 0.2$, leaving 0.4 in E only.



b(i) $1 - 0.4 = 0.6.$

b(ii) $0.3 + 0.2 = 0.5.$

c

$P(G')P(M) = 0.6(0.5) = 0.3$, while $P(G' \cap M) = 0.2$. They are dependent.

Exercise 2A – Statistics Y2 Pages 19-21

Solution 15

12. Events A and B are independent with $P(A) = x$ and $P(B) = y$. Find in terms of x, y:

(a) $P(A \cap B)$ (2)

(b) $P(A \cup B)$ (2)

(c) $P(A \cup B')$ (2)

a

$P(A \cap B) = P(A)P(B) = xy.$

b

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = x + y - xy.$

c

$P(A' \cap B') = 1 - (x + y - xy).$

Adding $P(A)$ gives $P(A \cup B') = 1 - y + xy.$

EXTENSION (Challenge Page 20)

Solution 16

Independent events A, B, C have probabilities x, y, z. Express in terms of x, y, z:

(a) $P(A \cap B \cap C)$

(b) $P(A \cup B \cup C)$

(c) $P((A \cup B') \cap C)$.

a

By independence, $P(A \cap B \cap C) = xyz.$

b

Add the individual probabilities, subtract pairwise intersections, then restore the triple intersection:

$x + y + z - xy - xz - yz + xyz.$

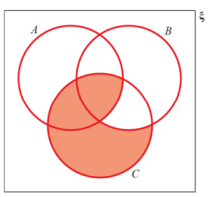
Alternatively, start with $P(A \cup B) = x + y - xy.$

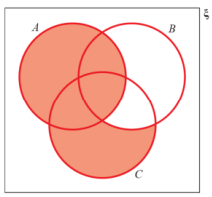
Its intersection with C has probability $xz + yz - xyz$. Add z and subtract this intersection.

c

The unshaded part of C has probability $P(C \cap B \cap A') = yz - xyz.$

Thus the required probability is $z - yz + xyz.$

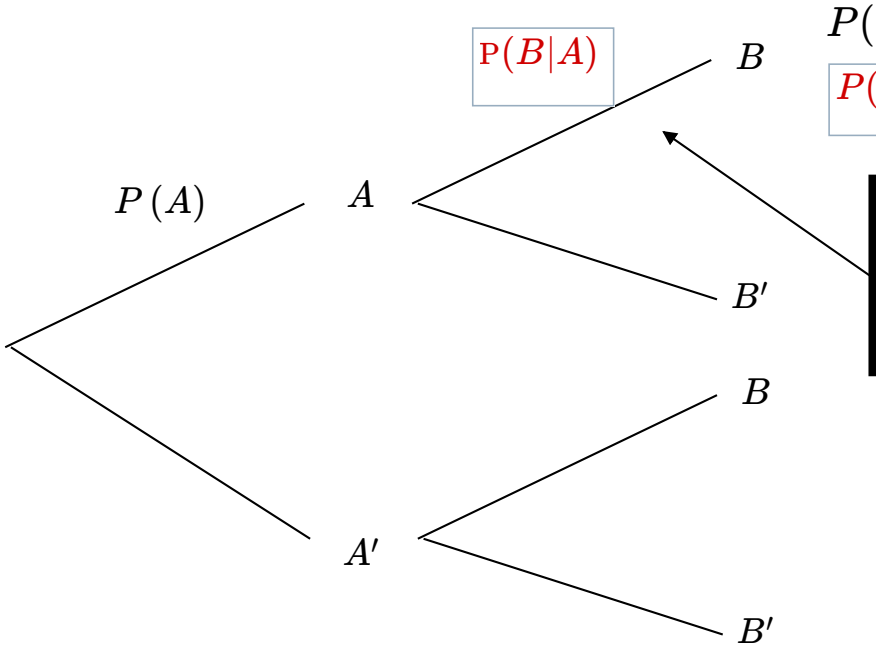




Conditional Probability

Solution 17

Think about how we formed a probability tree at GCSE:



$P(A \cap B) =$

$P(A)P(B \mid A)$

Read the '|' symbol as "**given that**". i.e. "B occurred **given that** A occurred".

Alternatively (and more commonly):

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

Memory Tip: You're dividing by the event you're conditioning on.

Examples

Solution 18

1. Year 9 language choices. B means the student is a boy. F means the student chose French.

	B	B'	Total
F	14	38	52
F'	26	22	48
Total	40	60	100

(a) Find $P(F \mid B')$ using both methods.

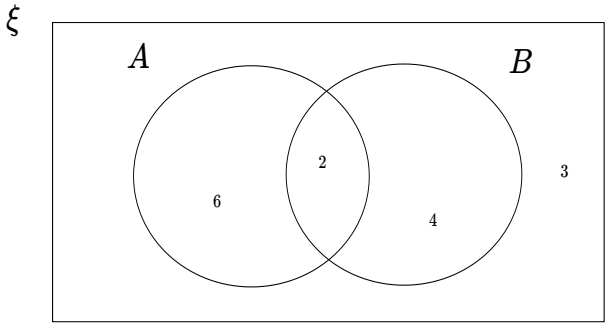
1a · Formula $P(F \mid B') = \frac{P(F \cap B')}{P(B')} = \frac{38/100}{60/100} = \frac{38}{60}.$

1a · Restricted sample Of the 60 students who are not boys, 38 study French. The probability is $38/60$.

(b) Find $P(B \mid F')$.

1b $P(B \mid F') = 26/48.$

2. Use the frequencies in the Venn diagram.



(a) Find $P(A \mid B)$ using both methods.

2a · Formula $P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{2/15}{6/15} = \frac{1}{3}.$

2a · Restricted sample Of the 6 outcomes in B, 2 are also in A. Hence $2/6 = 1/3$.

3

(b) $P(A' \mid B')$

2b $\frac{3}{9} = \frac{1}{3}$

(c) $P(B \mid A \cup B)$

2c $\frac{6}{12} = \frac{1}{2}$

Further Examples

Solution 19

a Given that $P(A) = 0.5$ and $P(A \cap B) = 0.3$, what is $P(B|A)$?

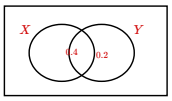
a

$$\mathbf{P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{0.3}{0.5} = 0.6}$$

b Given that $P(Y) = 0.6$ and $P(X \cap Y) = 0.4$, what is $P(X'|Y)$?
(Hint: Drawing a Venn Diagram will help!)

b

$\mathbf{P(X'|Y) = \frac{P(X' \cap Y)}{P(Y)}}$

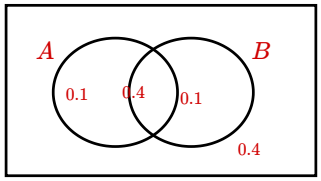


$\therefore \mathbf{P(X'|Y) = \frac{0.2}{0.6} = \frac{1}{3}}$

c Given that $P(A) = 0.5$, $P(B) = 0.5$ and $P(A \cap B) = 0.4$, what is $P(B|A')$?

c

$$\mathbf{P(B|A') = \frac{P(A' \cap B)}{P(A')} = \frac{0.1}{0.5} = 0.2}$$



Note: The 'restricted sample space' method also works for Venn Diagrams with probabilities.

Check your understanding

Solution 20

The events E and F satisfy $P(E) = 0.28$, $P(E \cup F) = 0.76$ and $P(E \cap F') = 0.11$. Find the following probabilities.

(a) $P(E \cap F) =$

a $0.28 - 0.11 = 0.17$

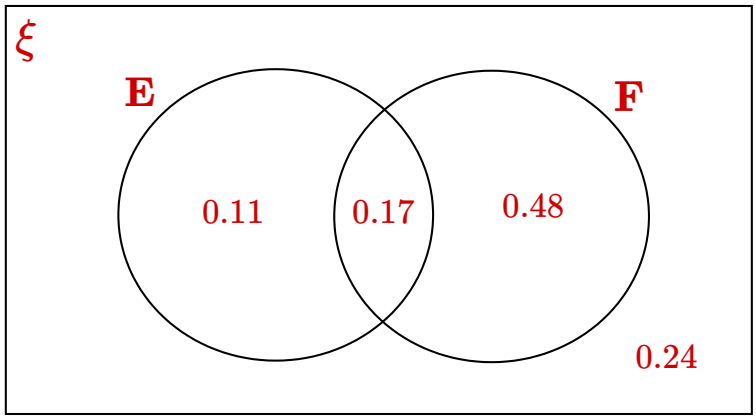
(b) $P(F) =$

b $0.76 - 0.28 + 0.17 = 0.65$

(c) $P(E' \mid F') =$

c $\frac{P(E' \cap F')}{P(F')} = \frac{0.24}{0.35} = \frac{24}{35}$

Venn diagram



Further Practice

Solution 21

1 $P(A \cap B') = 0.4, P(A \cup B) = 0.75$
Find $P(B)$, $P(A' \cap B')$.

1(i) $P(B) = 0.75 - 0.4 = 0.35$

1(ii) $P(A' \cap B') = 1 - 0.75 = 0.25$

2 $P(A) = 0.47$ and $P(A \cap B) = 0.12$ and $P(A' \cap B') = 0.03$
Find $P(A \mid B')$.

2
$$P(A \mid B') = \frac{P(A \cap B')}{P(B')} = \frac{0.47 - 0.12}{0.35 + 0.03} = \frac{35}{38}$$

3 $P(A') = 0.7, P(B') = 0.2, P(A \cap B') = 0.1$
Find $P(A \cup B')$, $P(B \mid A')$.

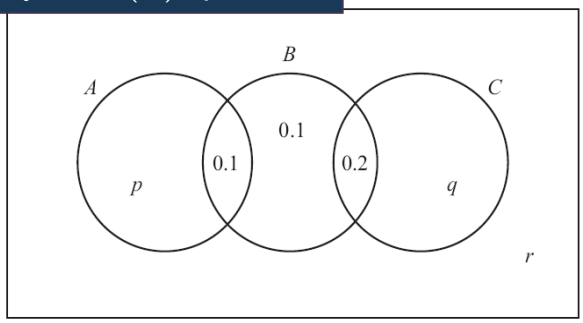
3(i) $P(A \cup B') = P(A) + P(B') - P(A \cap B') = 0.3 + 0.2 - 0.1 = 0.4$

3(ii) $P(B \mid A') = \frac{P(A' \cap B)}{P(A')} = \frac{1 - 0.4}{0.7} = \frac{6}{7}$

Further Test Your Understanding

Solution 22

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The Venn diagram in Figure 1 shows three events A, B and C and the probabilities associated with each region of B. The constants p, q and r each represent probabilities associated with the three separate regions outside B.

The events A and B are independent.

(a) Find the value of p. **(3)**

Given that $P(B|C) = \frac{5}{11}$,
(b) find the value of q and the value of r (4)
(c) Find $P(A \cup C|B)$ (2)

(a) **(From earlier)**
$0.1 = (p + 0.1) \times 0.4$
$p + 0.1 = 0.25$
$p = 0.15$

b
(b) $P(B|C) = \frac{P(B \cap C)}{P(C)}$
$\frac{5}{11} = \frac{0.2}{0.2 + q}$
$q = 0.24$
$r = 1 - 0.15 - 0.1 - 0.1 - 0.2 - 0.24$
$= 0.21$

c
(c) $P(A \cup C|B) = \frac{P((A \cup C) \cap B)}{P(B)}$
$= \frac{0.1 + 0.2}{0.4} = 0.75$

Exercise 2B – Statistics Y2 Pages 23-24 — Solution 23

8. Of 120 students, 56 watch sport S, 77 watch drama D, and 18 watch both.

(a) Draw a two-way table. (2)

(b) For a randomly chosen student, find:

(i) $P(D')$ (1)

(ii) $P(S' \cap D')$ (1)

(iii) $P(S \mid D)$ (1)

(iv) $P(D' \mid S)$ (1)

a

	D	D'	Total
S	18	38	56
S'	59	5	64
Total	77	43	120

Sport only: $56 - 18 = 38$. Drama only: $77 - 18 = 59$. Neither: $43 - 38 = 5$.

b(i) $\frac{43}{120}$.

b(ii) $\frac{5}{120} = \frac{1}{24}$.

b(iii) $\frac{18}{77}$.

b(iv) $\frac{38}{56} = \frac{19}{28}$.

Exercise 2B – Statistics Y2 Pages 23-24 — Solution 24

9. A rambling group has 63 women and 47 men. Of these, 26 women and 18 men use a walking stick.

(a) Draw a two-way table. (2)

(b) One rambler is selected at random. Find:

(i) $P(\text{uses a stick})$ (1)

(ii) $P(\text{uses a stick} \mid \text{female})$ (1)

(iii) $P(\text{male} \mid \text{uses a stick})$ (1)

a

	Women	Men	Total
Stick	26	18	44
No stick	37	29	66
Total	63	47	110

b(i) $44/110 = 2/5$.

b(ii) Of 63 women, 26 use a stick: $26/63$.

b(iii) Of 44 stick users, 18 are men: $18/44 = 9/22$.

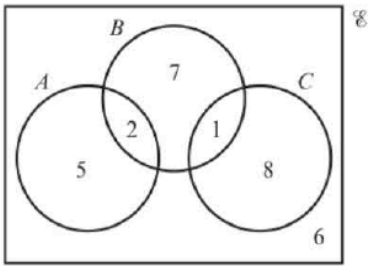
Exercise 2C – Statistics Y2 Pages 25-27 — Solution 25

7. The diagram shows students who watch TV programmes A, B, C. A student is selected at random. Given they watch at least one programme, find the probability they watch:

(a) programme C (2)

(b) exactly two programmes (2).

(c) Are watching B and watching C independent? Justify. (3)



a

Of 29 students, 23 watch at least one programme and 9 watch C. Hence $P(C \mid A \cup B \cup C) = \frac{9/29}{23/29} = \frac{9}{23} \approx 0.391$.

b

Exactly two programmes: $2 + 0 + 1 = 3$ students. The conditional probability is $\frac{3}{23} \approx 0.130$.

c

$P(B) = \frac{10}{29}$ and $P(B \mid C) = \frac{1/29}{9/29} = \frac{1}{9}$. These differ, so B, C are dependent.

Exercise 2C – Statistics Y2 Pages 25-27 — Solution 26

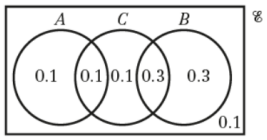
8. A, B are mutually exclusive and B, C independent. $P(A) = 0.2$, $P(B) = 0.6$, $P(C) = 0.5$ and $P(A' \cap B' \cap C') = 0.1$.

(a) Draw a Venn diagram. (4)

(b) Find (i) $P(A \mid C)$, (ii) $P(B \mid C')$, (iii) $P(C \mid A \cup B)$. (1 each)

a

$P(B \cap C) = 0.6(0.5) = 0.3$. Only B has probability 0.3. Hence $P(A \cup C) = 1 - 0.3 - 0.1 = 0.6$, giving $P(A \cap C) = 0.2 + 0.5 - 0.6 = 0.1$. The A-only and C-only regions each have probability 0.1.



b(i) $0.1/0.5 = 0.2$.

b(ii) $0.3/0.5 = 0.6$.

b(iii) $(0.1 + 0.3)/0.8 = 0.5$.

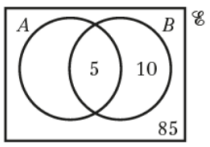
Exercise 2C – Statistics Y2 Pages 25-27

9. Of 100 people, 5 have an illness. All 5 test positive, as do 10 people without the illness. Let A denote having the illness and B testing positive.
(a) Draw a Venn diagram. (3)
(b) Calculate $P(A \mid B)$. (2)
(c) Use b to comment on the diagnostic test. (2)

Solution 27

a

All 5 people with the illness lie in the intersection. Ten more people test positive, and $100 - 5 - 10 = 85$ are in neither event.



b

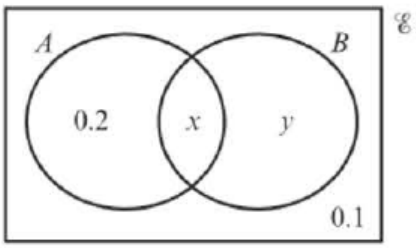
$P(A \mid B) = \frac{0.05}{0.15} = \frac{1}{3}$.

c

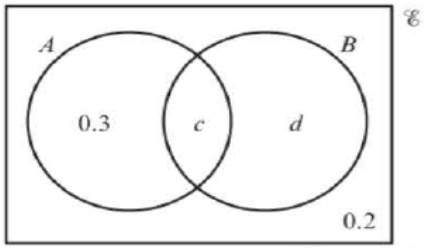
The test found everyone with the illness, but only one third of positive results were true positives in this study.

Exercise 2C – Statistics Y2 Pages 25-27

11. Given $P(A \mid B) = P(B')$, use the diagram to find x, y.



12. Given $P(A \mid B) = P(A')$, use the diagram to find c, d.



Solution 28

11

$x + y = 1 - 0.2 - 0.1 = 0.7$.
$\frac{x}{x+y} = P(B') = 0.3$.
$x = 0.3(0.7) = 0.21$ and $y = 0.7 - 0.21 = 0.49$.

12

$c + d = 1 - 0.3 - 0.2 = 0.5$.
$\frac{c}{c+d} = d + 0.2$, so $c = 0.5d + 0.1$.
$1.5d + 0.1 = 0.5$ gives $d = \frac{4}{15}$ and $c = \frac{7}{30}$.

Extension

Solution 29

I have two children. At least one is a boy. What is the probability that both are boys? Assume independent births and equally likely sexes.

The restricted sample space is $\{BB, BG, GB\}$. Only BB satisfies the condition that both are boys, giving probability $\frac{1}{3}$.

Full Laws of Probability

Solution 30

If events A and B are independent.

$$P(A \cap B) = P(A)P(B)$$
$$P(A \mid B) = P(A)$$

If events A and B are mutually exclusive:

$$P(A \cap B) = 0$$
$$P(A \cup B) = P(A) + P(B)$$

In general:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

We first encountered this in the previous section.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

This is known as the **Addition Law.**

Informal Proof: If we added the probabilities in the A and B sets in the Venn Diagram, we'd be double counting the intersection, so subtract so that it's only counted once.

Example

Solution 31

6. Edexcel S1
(a) Explain what a sample space is. (1)
(b) Explain what an event is. (1)
A and B are independent with $P(A) = \frac{1}{3}$ and $P(B) = \frac{1}{4}$. Find:
(c) $P(A \cap B)$. (1)
(d) $P(A \mid B)$. (2)
(e) $P(A \cup B)$. (2)

a
The set of all possible outcomes.

b
An event is a subset of the sample space.

c
$P(A \cap B) = \frac{1}{3} \times \frac{1}{4} = \frac{1}{12}$.

d
$P(A \mid B) = P(A) = \frac{1}{3}$ by independence.

e
$P(A \cup B) = \frac{1}{3} + \frac{1}{4} - \frac{1}{12} = \frac{1}{2}$.

Further Examples

Solution 32

Textbook. $P(C) = 0.2$, $P(D) = 0.6$ and $P(C \mid D) = 0.3$. Find:
(a) $P(C \cap D)$
(b) $P(D \mid C)$
(c) $P(C \cup D)$

a
$0.3 = \frac{P(C \cap D)}{0.6}$, so $P(C \cap D) = 0.18$.

b
$P(D \mid C) = \frac{0.18}{0.2} = 0.9$.

c
$P(C \cup D) = 0.2 + 0.6 - 0.18 = 0.62$.

10. Jan 2012 Q2. (a) Describe the relationship between R and S when $P(R \cap S) = 0$. (1)
A, B are independent, $P(A) = \frac{1}{4}$ and $P(A \cup B) = \frac{2}{3}$. Find:
(b) $P(B)$ (4) (c) $P(A' \cap B)$ (2)
(d) $P(B' \mid A)$ (2)

a They are mutually exclusive.

b
$\frac{2}{3} = \frac{1}{4} + P(B) - \frac{1}{4}P(B)$ by addition and independence.
$\frac{5}{12} = \frac{3}{4}P(B)$, giving $P(B) = \frac{5}{9}$.

c $P(A' \cap B) = \frac{3}{4} \times \frac{5}{9} = \frac{5}{12}$.

d
$P(B' \mid A) = P(B') = 1 - \frac{5}{9} = \frac{4}{9}$.

Hint for 10(b). Independence gives $P(A \cap B) = P(A)P(B)$. Knowing $P(A \cup B)$ suggests the addition rule.

Test Your Understanding

Solution 33

Edexcel S1

9. Events A, B, C are defined in sample space S. A and B are mutually exclusive. A and C are independent.

(a) Draw a Venn diagram showing the three events and the sample space. (3)

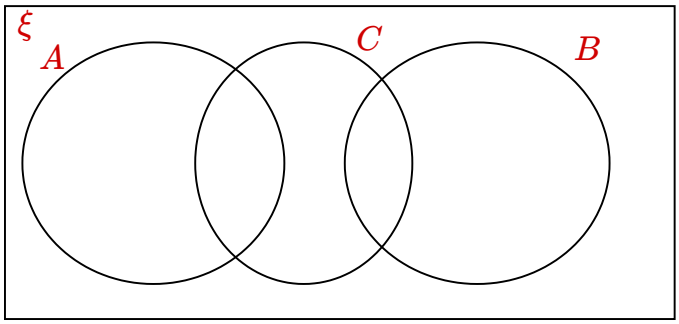
Given $P(A) = 0.2$, $P(B) = 0.4$ and $P(A \cup C) = 0.7$, find:

(b) $P(A \mid C)$ (2)

(c) $P(A \cup B)$ (2)

(d) $P(C)$ (4)

a



b $P(A \mid C) = P(A) = 0.2.$

c $P(A \cup B) = 0.2 + 0.4 = 0.6.$

d

$0.7 = 0.2 + P(C) - 0.2P(C).$

$0.5 = 0.8P(C)$, so $P(C) = \frac{5}{8}.$

SUPER IMPORTANT TIPS

Solution 34

If I were to identify two tips that will possible help you the most in probability questions:

If you see the words 'given that', Immediately write out the law for conditional probability.

Example: "Given Bob walks to school, find the probability that he's not late..."

First thing you should write $P(L' \mid W) = \frac{P(L' \cap W)}{P(W)} = \ldots$

If you see the words 'are independent', Immediately write out the laws for independence.
(Even before you've finished reading the question!)

Example: "A is independent from B..."

First thing you should write

$P(A \cap B) = P(A)P(B)$

$P(A \mid B) = P(A)$

If you're stuck on a question where you have to find a probability given others, it's probably because you've failed to take into account that two events are independent or mutually exclusive, or you need to use the conditional probability or additional law.

Exercise 2D – Statistics Y2 Pages 29-30

Solution 35

9. $P(A) = 0.42$, $P(B) = 0.37$, $P(A \cap B) = 0.12$. Find:

(a) $P(A \cup B)$ (2)

(b) $P(A \mid B')$ (2)

$P(C) = 0.3$; B, C are mutually exclusive and A, C independent.

(c) Find $P(A \cap C)$. (2)

(d) Draw a Venn diagram showing all regions. (4)

(e) Find $P((A' \cup C)')$. (2)

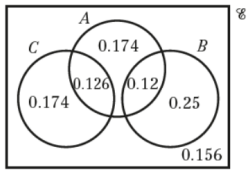
a $0.42 + 0.37 - 0.12 = 0.67$.

b $\frac{0.42-0.12}{1-0.37} = 0.476$ (3 s.f.).

c $0.42(0.3) = 0.126$.

d

C only: $0.3 - 0.126 = 0.174$. B only: $0.37 - 0.12 = 0.25$. A only: $0.42 - 0.12 - 0.126 = 0.174$. The outside region is $1 - 0.844 = 0.156$.



e $P(A' \cup C) = 0.174 + 0.126 + 0.25 + 0.156 = 0.706$. Its complement has probability 0.294.

Exercise 2D – Statistics Y2 Pages 29-30

Solution 36

10. $P(A) = 0.4$, $P(B) = 0.7$, $P(C) = 0.4$ and $P(A \cap B) = 0.3$. Events A, C are mutually exclusive and B, C independent. Find each probability. (1 each)

(a) $P(B \cap C)$

(b) $P(B \mid C)$

(c) $P(A \mid B')$

(d) $P(B \cap C \mid A')$

a Independence gives $0.7(0.4) = 0.28$.

b $0.28/0.4 = 0.7$.

c $\frac{0.4-0.3}{1-0.7} = \frac{1}{3}$.

d $A \cap B \cap C$ is empty, so the numerator is 0.28. The denominator is $1 - 0.4 = 0.6$, giving 0.467 (3 s.f.).

Exercise 2D – Statistics Y2 Pages 29-30

Solution 37

11. A and B are the events that Anna and Bella are late for school. $P(A) = 0.3$, $P(B) = 0.7$, and $P(A' \cap B') = 0.1$. Find the probability that:

(a) both are late (1)

(b) Anna is late given that Bella is late. (2)

(c) Their teacher suspects their lateness is linked. Comment and show your working. (2)

a

$P(A \cup B) = 0.9$, so $P(A \cap B) = 0.3 + 0.7 - 0.9 = 0.1$.

b

$0.1/0.7 = 0.143$ (3 s.f.).

c

$P(A)P(B) = 0.21 \neq 0.1 = P(A \cap B)$. They are dependent. Each is less likely to be late when the other is late.

Exercise 2D – Statistics Y2 Pages 29-30

Solution 38

12. John and Kayleigh play darts on the same team. Let J, K denote their respective wins. $P(J) = 0.6$, $P(K) = 0.7$ and $P(J \cup K) = 0.8$. Find the probability that:

(a) both win (1)

(b) John wins given Kayleigh loses (2)

(c) Kayleigh wins given John wins. (2)

(d) Are J, K independent? Show your working. (2)

a

$0.6 + 0.7 - 0.8 = 0.5$.

b

$P(J \mid K') = \frac{0.6-0.5}{1-0.7} = \frac{1}{3}$.

c

$P(K \mid J) = 0.5/0.6 = 5/6$.

d

$P(K \mid J) = 5/6 \neq 0.7 = P(K)$. They are dependent.

Extension (Challenge Page 30) — Solution 39

$P(X = x) = kx$ for $x = 1, 2, 3, 4, 5$.

(a) Find k.

(b) Find $P(X = 5 \mid X > 2)$.

(c) Find the probability that X is odd given that it is prime.

a

$k + 2k + 3k + 4k + 5k = 1$, so $k = 1/15$.

b

$$\frac{5/15}{1-3/15} = \frac{5}{12}.$$

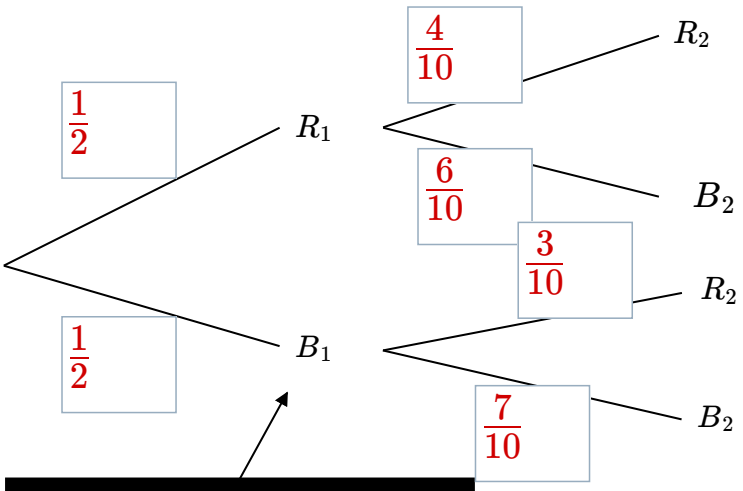
c

The prime outcomes are 2, 3 and 5. Of these, 3 and 5 are odd, so the probability is $\frac{3+5}{2+3+5} = \frac{4}{5}$.

Probability Trees — Solution 40

Probability Trees can be used to **determine a conditional probability.**

Example: You have two bags, the first with 5 red balls and 5 blue balls, and the second with 3 red balls and 6 blue balls. You first pick a ball from the first bag, and place it in the second. You then pick a ball from the second bag. Complete the tree diagram.



(a) Find the probability of red on the second pick.

a

$P(R_2) = P(R_1 \cap R_2) + P(B_1 \cap R_2)$
$= \frac{1}{5} + \frac{3}{20} = \frac{7}{20}$.

(b) Given red on the second pick, find the probability of red on the first.

b

$P(R_1 \mid R_2) = \frac{P(R_1 \cap R_2)}{P(R_2)}$
$= \frac{1/5}{7/20} = \frac{4}{7}$.

Further Example

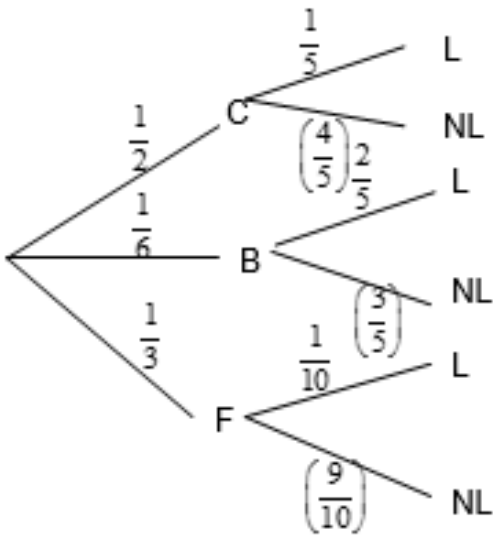
Solution 41

Edexcel S1 May 2009 Q2. Bill travels by car, bicycle or foot with probabilities $\frac{1}{2}, \frac{1}{6}, \frac{1}{3}$. The probabilities he is late when using each method are $\frac{1}{5}, \frac{2}{5}, \frac{1}{10}$ respectively.

(a) Draw a probability tree.

(b) Given Bill is late, find the probability he did not travel on foot. (4)

a



b

$$P(F' \mid L) = \frac{P(F' \cap L)}{P(L)}.$$

$$P(F' \cap L) = \frac{1}{2} \times \frac{1}{5} + \frac{1}{6} \times \frac{2}{5} = \frac{1}{6}.$$

$$P(L) = \frac{1}{6} + \frac{1}{3} \times \frac{1}{10} = \frac{1}{5}.$$

Hence $P(F' \mid L) = \frac{1/6}{1/5} = \frac{5}{6}.$

Testing Your Understanding

Solution 42

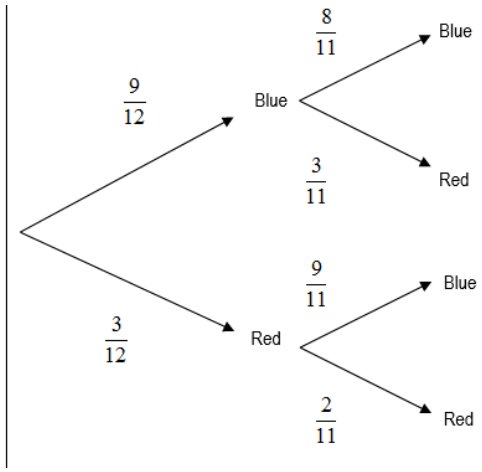
6. Edexcel S1, Jan 2006 Q4. A bag contains 9 blue and 3 red balls. A ball is selected at random and its colour recorded. It is not replaced. A second ball is selected at random and its colour recorded.

(a) Draw a probability tree. (3)

(b) Find the probability the second ball is red. (2)

(c) Given the second ball is red, find the probability both balls are red. (2)

a



b

$$P(R_2) = \frac{9}{12} \times \frac{3}{11} + \frac{3}{12} \times \frac{2}{11} = \frac{1}{4}.$$

c

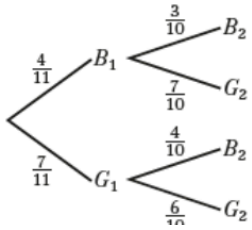
$$P(R_1 \mid R_2) = \frac{\frac{3}{12} \times \frac{2}{11}}{1/4} = \frac{2}{11}.$$

Exercise 2E – Statistics Y2 Pages 31-34 — Solution 43

6. Two balls are drawn at random without replacement from a bag of 4 blue and 7 green balls.
(a) Draw a probability tree. (3)
(b) Find the probability the second ball is green. (2)
(c) Given that the second ball is green, find the probability both are green. (2)

a

The first draw leaves 10 balls. Each second-draw branch depends on the colour removed.



b

$$\frac{4}{11}\frac{7}{10} + \frac{7}{11}\frac{6}{10} = \frac{7}{11}.$$

c

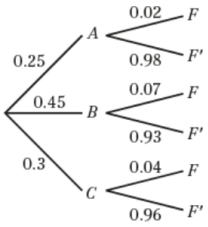
$$\frac{(7/11)(6/10)}{7/11} = \frac{3}{5} = 0.6.$$

Exercise 2E – Statistics Y2 Pages 31-34 — Solution 44

7. Factories A, B and C produce tin sheets. A supplies 25%, B supplies 45% and C supplies the rest. Their flaw probabilities are 0.02, 0.07 and 0.04.
(a) Draw a probability tree. (3)
(b) Find the probability a random sheet is (i) from B and flawed (2), (ii) flawed (3).
(c) Given a flawed sheet, find the probability it is from A. (3)

a

C supplies 30%. For each factory, the probability of no flaw is one minus its flaw probability.



b(i)

$$0.45(0.07) = 0.0315.$$

b(ii)

$$0.25(0.02) + 0.0315 + 0.30(0.04) = 0.0485.$$

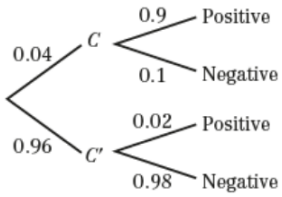
c

$$\frac{0.25(0.02)}{0.0485} = 0.103 \text{ (3 s.f.)}.$$

8. A genetic condition affects 4% of a population. The probability of a positive test is 0.9 for those with the condition and 0.02 for those without it.
(a) Draw a probability tree. (3)
(b) Find the probability a random person tests negative. (3)
(c) Given a negative result, find the probability the person has the condition. (2)
(d) Comment on the test's effectiveness. (1)

a

Split first by condition, then by test result.



b

$0.04(0.10) + 0.96(0.98) = 0.9448.$

c

$\frac{0.04(0.10)}{0.9448} = 0.00423$ (3 s.f.).

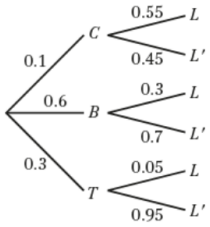
d

The test misses 10% of people with the condition. The fraction of positive results that are false is $0.96(0.02)/(1 - 0.9448) = 0.348$, over one third.

9. Bill travels to work by car, bus or train with probabilities 0.1, 0.6 and 0.3. The probabilities he is late with these methods are 0.55, 0.3 and 0.05 respectively.
(a) Draw a probability tree. (3)
(b) Find the probability (i) Bill travels by train and is late (2), (ii) Bill is late (2).
(c) Given Bill is late, find the probability he did not travel by car. (4)

a

For each method, the probabilities of late and on time sum to 1.



b(i)

$0.3(0.05) = 0.015.$

b(ii)

$0.1(0.55) + 0.6(0.3) + 0.3(0.05) = 0.25.$

c

$\frac{0.6(0.3)+0.3(0.05)}{0.25} = 0.78.$

Exercise 2E – Statistics Y2 Pages 31-34 Solution 47

11. A box contains 7 sweet and 3 sour jelly beans. Two are selected and eaten. Emilia wants the probability both are sweet given that at least one is sweet.

She calculates $P(\text{both sweet}) = (7/10)^2 = 49/100$ and $P(\text{at least one sweet}) = 1 - (3/10)^2 = 91/100$. Her answer is $49/91$.

Identify her mistake and find the correct probability. (4)

11

The first bean is eaten, so only 9 remain. If it was sweet, the probability of a second sweet is $6/9$.

$P(\text{both sweet}) = \frac{7}{10}\frac{6}{9} = \frac{7}{15}$.

$P(\text{at least one sweet}) = 1 - \frac{3}{10}\frac{2}{9} = \frac{14}{15}$.

The required probability is $(7/15)/(14/15) = 1/2$.

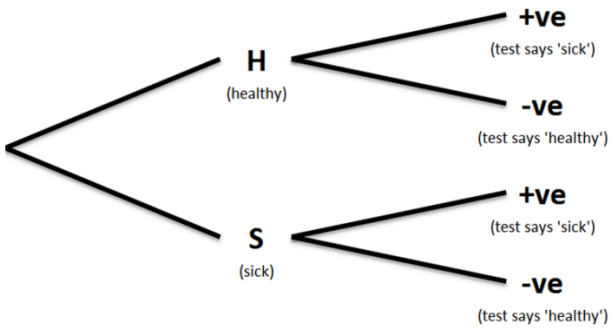
Appendix - Problem solving Example 1 Solution 48

Almost all medical tests have a small chance of a 'false positive' or a 'false negative'.

- **False positive**: The test says you **do** have the condition when you really **don't**. (you're fine, but the test thinks you're sick)
- **False negative**: The test says you **don't** have the condition when you really **do**. (you're sick, but the test thinks you're fine)

- The test for a particular disease has a false positive rate of 5%.
  *This means that 5% of people who **don't** have the disease will be told that they **do**.*
- The test has a false negative rate of 1%.
  *This means that 1% of people who **do** have the disease will be told that they **don't**.*

Assuming this particular disease affects $10\%$of people, complete the tree diagram below:



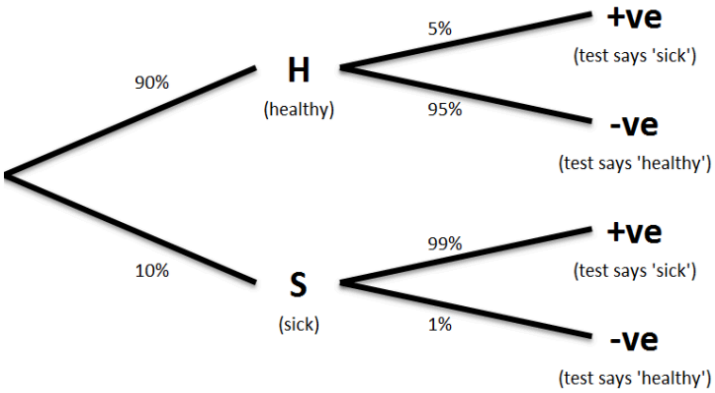
Appendix - Problem solving Example 1

Use the probability tree to calculate the four outcomes.

	Healthy	Sick
Positive		
Negative		

A positive result for someone healthy is a false positive and can lead to unnecessary treatment or worry. A negative result for someone sick is a false negative and can delay treatment.
If the test says a person is sick, what is the probability they really are? Hint: imagine 1000 people and count the sick people among all positive tests.

Appendix - Example 1 Solution



Multiply the branch probabilities.
$0.9(0.05) = 0.045$ and $0.9(0.95) = 0.855$.
$0.1(0.99) = 0.099$ and $0.1(0.01) = 0.001$.

	Healthy	Sick
Positive	4.5%	9.9%
Negative	85.5%	0.1%

In 1000 people, $45 + 99 = 144$ test positive. Of these, 99 really are sick.
$99/144 = 0.6875$, giving a probability of 68.75%.

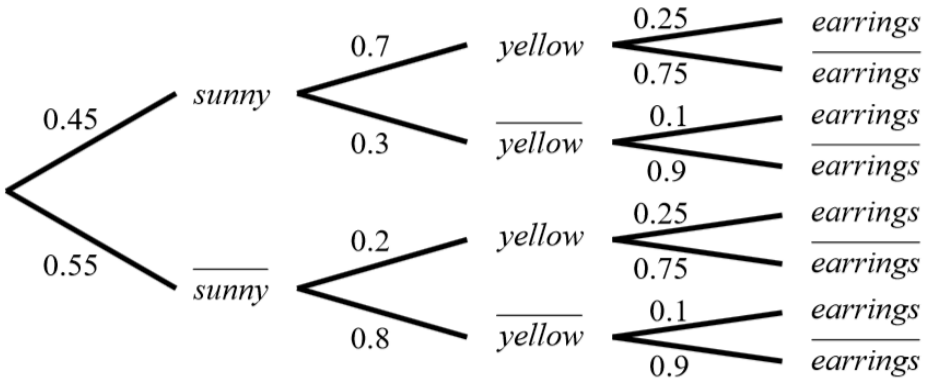
Appendix - Problem solving Example 2

Solution 51

Mary chooses her outfit based on the weather. The probability it is sunny is 0.45. She wears yellow with probability 0.7 when it is sunny and 0.2 otherwise. If she wears yellow, she wears her favourite earrings with probability 0.25; otherwise the probability is 0.1.
Given that Mary is not wearing her favourite earrings, find the probability she is wearing yellow.

Appendix - Example 2 Solution

Solution 52



Let Y denote yellow and E her favourite earrings.

$P(Y \cap E') = 0.45(0.7)(0.75) + 0.55(0.2)(0.75) = 0.31875.$

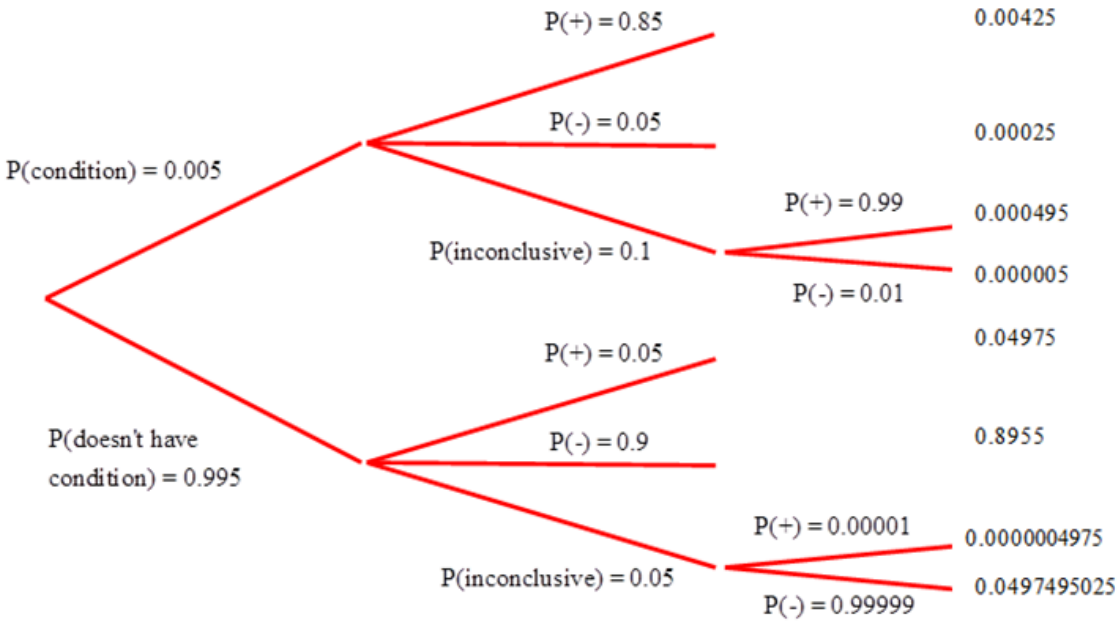
$P(E') = 0.23625 + 0.1215 + 0.0825 + 0.396 = 0.83625.$

$P(Y \mid E') = \frac{0.31875}{0.83625} = 0.381$ (3 s.f.).

Appendix - Problem solving Example 3

A condition affects 0.5% of a population. A test can be positive, negative or inconclusive.
Of patients with the condition, 85% test positive, 5% negative and 10% inconclusive. Of those without it, 5% test positive, 90% negative and 5% inconclusive.
Inconclusive tests are followed by a thorough examination. The examination detects the condition in 99% of patients who have it. It reports the condition in 0.001% of patients who do not have it.
Given that a patient does not have the condition, find the probability of a final positive result.

Appendix - Example 3 Solution



$$P(\text{positive} \mid \text{no condition}) = \frac{0.04975 + 0.0000004975}{0.995}$$
$$= 0.0500005.$$